

Project MTH686

Non-Linear Regression Analysis

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Introduction:

In this project, we will work with three non-linear regression models.

The models are:

Model 1:

$$y(t) = \alpha_0 + \alpha_1 e^{\beta_1 t} + \alpha_2 e^{\beta_2 t} + \epsilon(t)$$

Model 2:

$$y(t) = \frac{\alpha_0 + \alpha_1 t}{\beta_0 + \beta_1 t} + \epsilon(t)$$

Model 3:

$$y(t) = \alpha_0 + \alpha_1 t + \alpha_2 t^2 + \alpha_3 t^3 + \alpha_4 t^4 + \epsilon(t)$$

Questions:

1. Find the least squares estimators of the unknown parameters under three different model assumptions.

Sol: The least squares estimators (LSE) are the values of the model parameters that minimize the sum of squared residuals, where the residuals are the differences between the observed data points and the predicted values from the model.

The least square estimators for each model:

Model 1:

$$y(t) = \alpha_0 + \alpha_1 e^{\beta_1 t} + \alpha_2 e^{\beta_2 t} + \epsilon(t)$$

The least squares estimators for the parameters $\alpha_0, \alpha_1, \alpha_2, \beta_0$ and β_1 are:

$$\alpha_0 = -89.586, \alpha_1 = 93.845, \alpha_2 = 4.064, \beta_1 = 0.0109, \beta_1 = 1.591$$

Model 2:

$$y(t) = \frac{\alpha_0 + \alpha_1 t}{\beta_0 + \beta_1 t} + \epsilon(t)$$

The least squares estimators for the parameters $\alpha_0, \alpha_1, \beta_0$ and β_1 are:

$$\alpha_0 = 0.206, \alpha_1 = 0.103, \beta_0 = 0.025, \beta_1 = -0.013$$

Model 3:

$$y(t) = \alpha_0 + \alpha_1 t + \alpha_2 t^2 + \alpha_3 t^3 + \alpha_4 t^4 + \epsilon(t)$$

The least squares estimators for the parameters $\alpha_0, \alpha_1, \alpha_2, \alpha_3$ and α_4 are:

$$\alpha_0 = 8.314, \alpha_1 = 7.600, \alpha_2 = 4.893, \alpha_3 = 2.526, \alpha_4 = 1.894$$

2. How did you find the least squares estimators? What kind of initial guesses you have chosen?

Sol: In models 1 and 2, I used Linear model (lm) method to find the coefficients.

In Model 1, The initial guesses for $\alpha_0, \alpha_1, \alpha_2, \beta_0$ and β_1 are:

$$\alpha_0 = 1, \alpha_1 = 1, \alpha_2 = 1, \beta_1 = -0.1, \beta_1 = -0.1$$

The values of α_0 and α_1 are set to 1, assuming both exponential terms start with similar contribution. The negative values for β_0 and β_1 are chosen because exponential decay is often observed in many real-world systems.

In Model 2, The initial guesses for $\alpha_0, \alpha_1, \beta_0$ and β_1 are:

$$\alpha_0 = 1, \alpha_1 = 1, \beta_1 = 1, \beta_1 = 1$$

Here, the initial guesses are set to 1 for all parameters, assuming that the rational model's behaviour is expected to follow a relatively simple form where all terms

contribute similarly. These initial guesses provide a baseline to start the optimization process.

In Model 3, the parameters $\alpha_0, \alpha_1, \alpha_2, \alpha_3$ and α_4 are estimated using least squares method. Since this model is polynomial, no explicit initial guesses were provided. Instead, the least squares approach directly estimates the coefficients based on the data. This approach assumes that the model's form (a polynomial) is the correct one and the coefficients are found by solving the normal equations of the least squares problem.

3. Find the best fitted model.

Sol: To find the best-fitted model, we can evaluate each model's goodness of fit using metrics, RSS (Residual Sum of Squares), R^2 (Coefficient of Determination), AIC (Akaike Information Criterion), and BIC (Bayesian Information Criterion). These metrics help us determine which model best fits the data.

Model	RSS	R^2	AIC	BIC
Model 1	1.59	0.9991	-279.14	-267.55
Model 2	0.65	0.9996	-348.68	-339.41
Model 3	0.47	0.9997	-317.08	-359.49

In general, Model having low AIC is the best model. Therefore, Model 3 is the best model. Furthermore, models having lower RSS, Higher R^2 and lower BIC value are preferred. All these points are true for Model 3.

4. Find the estimate of σ^2 .

Sol: The formula for σ^2 in terms of RSS is:

$$\sigma^2 = \frac{RSS}{n - p}$$

Where n and p are number of data points and number of parameters in the model respectively.

Given data has 75 data points and p for Model 1, Model 2 and model 3 is 5,4 and 5 respectively.

Therefore, estimated values of σ^2 are:

Model 1: 0.0227

Model 2: 0.0091

Model 3: 0.0067

5. Find the associated confidence intervals based on the Fisher information matrix.

Sol: Based on the Fisher Information Matrix and the second derivatives of the log-likelihood, we compute the following 95% confidence intervals for each model:

Model 1:

- Parameter 1 (α_0): (29397.3818, 648863.7342)
- Parameter 2 (α_1): (-836401.3575, -216984.1948)
- Parameter 3 (α_2): (187546.2430, 187595.4546)
- Parameter 4 (β_1): (-36.3205, 36.3103)
- Parameter 5 (β_2): (-2.4837, 2.4552)

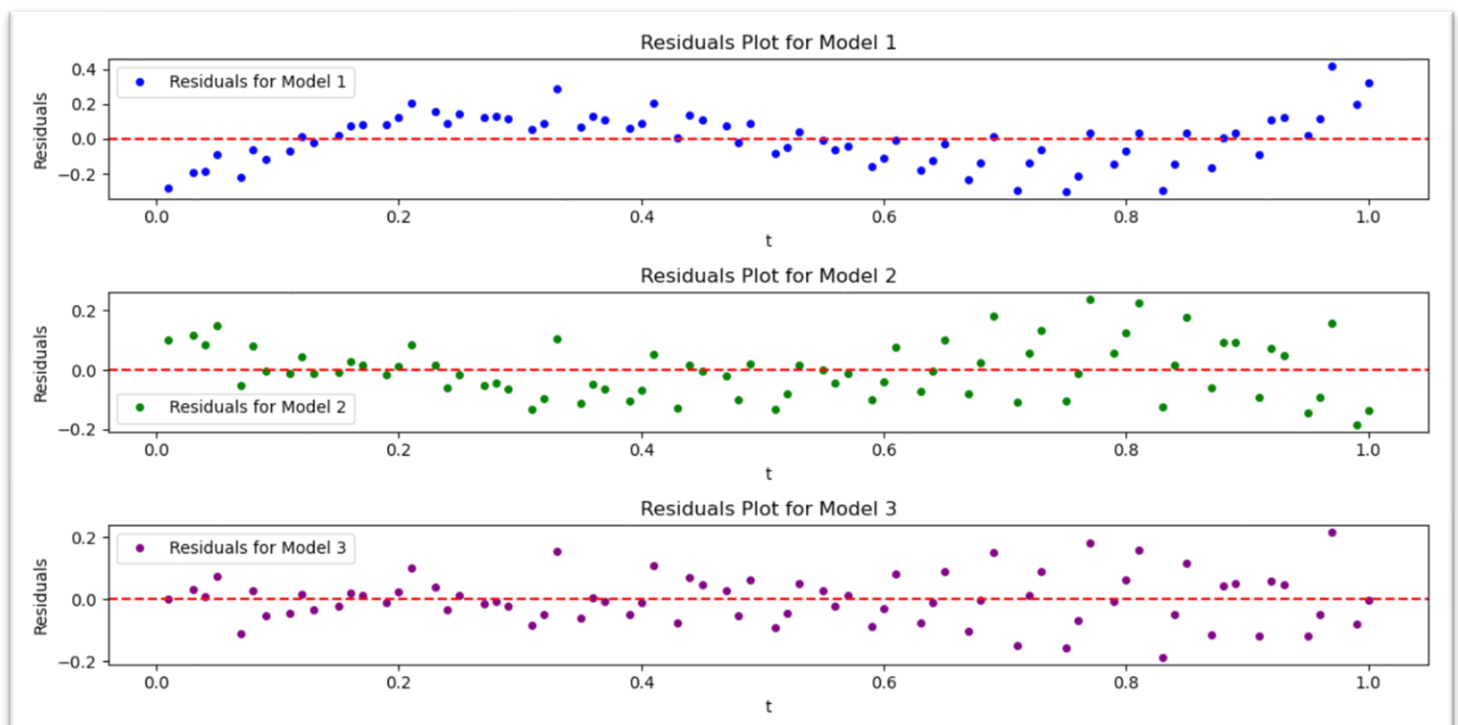
Model 2:

- Parameter 1 (α_0): (-19328.0503, 19328.4632)
- Parameter 2 (α_1): (-9668.2796, 9668.4861)
- Parameter 3 (β_0): (-2354.9599, 2355.0102)
- Parameter 4 (β_1): (-1211.6823, 1211.6564)

Model 3 (Polynomial Model):

- Parameter 1 (α_0): (8.2146, 8.4137)
- Parameter 2 (α_1): (6.2586, 8.9415)
- Parameter 3 (α_2): (-0.4419, 10.2285)
- Parameter 4 (α_3): (-5.3619, 10.4141)
- Parameter 5 (α_4): (-1.9688, 5.7570)

6. Plot the residuals.



7. Test whether it satisfies the normality assumption or not?

Sol:

Model 1:

- Shapiro-Wilk Test: $W=0.9834$, $p\text{-value} = 0.4310$.
- Since the p-value (0.4310) is greater than 0.05, we fail to reject the null hypothesis, meaning the residuals of Model 1 appear to follow a normal distribution.

Model 2:

- Shapiro-Wilk Test: $W=0.9774$, $p\text{-value} = 0.1998$.
- Since the p-value (0.1998) is greater than 0.05, we fail to reject the null hypothesis, meaning the residuals of Model 2 appear to follow a normal distribution.

Model 3:

- Shapiro-Wilk Test: $W=0.9912$, $p\text{-value} = 0.8887$.
- Since the p-value (0.8887) is greater than 0.05, we fail to reject the null hypothesis, meaning the residuals of Model 3 appear to follow a normal distribution.

8. Plot the observed data points and fitted curve.

